

How to shift unproductive *Kalmia angustifolia* – *Rhododendron groenlandicum* heath to productive conifer plantation¹

Nelson Thiffault and Robert Jobidon

Abstract: Conifer-regeneration failure is often observed on sites invaded by ericaceous shrubs. In northeastern Quebec, Canada, these sites are frequently characterized by dense *Kalmia angustifolia* L. – *Rhododendron groenlandicum* (Oeder) K.A. Kron & Judd cover. Such failures are potential consequences of nutrient limitation, allelopathy, or low soil temperatures. Conversion of productive forests into heaths poses a threat to the maintenance of forest productivity and biodiversity. We evaluated scarification, spot fertilization, and increased seedling foliar N concentration as treatments to promote planted black spruce (*Picea mariana* (Mill.) BSP) seedling survival and growth. We measured seedling, vegetation, and soil responses to the treatments for 5 years following planting. Scarification had positive impacts on seedling growth: the differences between scarified and unscarified plots increased over time, and double-pass scarification proved slightly more effective than a single-pass treatment. Responses to scarification were enhanced when seedlings were fertilized. A slow-release fertilizer with micronutrients proved slightly more effective than the 26N–12P–6K formulation; the latter also induced higher mortality than the former or no fertilizer. Gains due to increased N concentrations based on nursery practices were significant but short-lived. Our results demonstrate how silviculture and nursery practices can be used for resetting the secondary succession where ecosystem retrogression is observed following *K. angustifolia* – *R. groenlandicum* invasion.

Résumé : Des échecs de régénération coniférienne sont souvent observés sur des stations envahies par les éricacées. Dans le nord-est du Québec (Canada), ces stations sont fréquemment caractérisées par un couvert dense de *Kalmia angustifolia* L. – *Rhododendron groenlandicum* (Oeder) K.A. Kron & Judd. Ces échecs sont la conséquence possible d'interférences nutritionnelles, d'allélopathie, ou de basses températures du sol. La conversion de forêts productives en landes menace la productivité forestière et la biodiversité. Nous avons évalué comment le scarifiage, la fertilisation et l'utilisation de plants à fortes concentrations foliaires en N pouvaient favoriser la croissance et la survie de plants d'épinette noire (*Picea mariana* (Mill.) BSP). Nous avons mesuré les réponses des plants, de la végétation et du sol pendant cinq ans suivant la mise en terre. Le scarifiage a eu des effets positifs sur la croissance : les différences entre les parcelles scarifiées et témoins ont augmenté dans le temps et le scarifiage à double passages a été un peu plus efficace que celui à simple passage. Les réponses au scarifiage ont été améliorées par la fertilisation. L'engrais à libération lente avec micronutriments a été légèrement plus efficace que la formulation 26N–12P–6K; cette dernière a également augmenté la mortalité par rapport à la première et au témoin. Les gains associés aux concentrations élevées de N foliaire (issues des pratiques de pépinière) ont été significatifs mais de courte durée. Nos résultats illustrent comment la sylviculture et les pratiques de pépinière peuvent être utilisées pour redémarrer la succession secondaire, là où une rétrogression de l'écosystème est observée suite à l'envahissement par *K. angustifolia* – *R. groenlandicum*.

Introduction

In some ecosystems, conifer regeneration may be hindered by the presence of ericaceous species following fire or harvesting (Mallik 2003). Examples of failure of forest tree regeneration due to the presence of ericaceous shrubs are

found in Sweden (Nilsson and Wardle 2005), Scotland (Mallik 1995), and the northeastern United States (Latham 2003). In Canada this phenomenon is frequently observed in western coastal forests of British Columbia (Prescott et al. 1996), as well as in eastern temperate and boreal forests of Newfoundland, Nova Scotia, and Ontario (Anonymous

Received 9 November 2005. Accepted 7 April 2006. Published on the NRC Research Press Web site at <http://cjfr.nrc.ca> on 30 September 2006.

N. Thiffault.² Direction de la recherche forestière, Ministère des Ressources naturelles et de la Faune du Québec, 2700 rue Einstein, Québec, QC G1P 3W8, Canada, and Adjunct Professor, Faculté de foresterie et de géomatique, Université Laval, QC G1K 7P4, Canada, and Faculté des sciences, Université de Sherbrooke, QC J1K 2R1, Canada.

Robert Jobidon. Direction de la recherche forestière, Ministère des Ressources naturelles et de la Faune du Québec, 2700 rue Einstein, Québec, QC G1P 3W8, Canada.

¹This article is one of a selection of papers published in the Special Issue on Forest Vegetation Management.

²Corresponding author (e-mail: nelson.thiffault@mrnf.gouv.qc.ca).

1972; Titus et al. 1995; Wall 1977). In Quebec, transformation of once productive, relatively diverse forest stands into heaths dominated by sheep laurel, *Kalmia angustifolia* L., or bog Labrador tea, *Rhododendron groenlandicum* (Oeder) K.A. Kron & Judd, following harvesting or wildfire is considered a significant forestry issue (Grondin and Cimon 2003; Jobidon 1995). The rapid and lasting domination of disturbed sites by ericaceous shrubs, and the associated stunted growth of conifers, pose a threat to the maintenance of forest productivity and biodiversity.

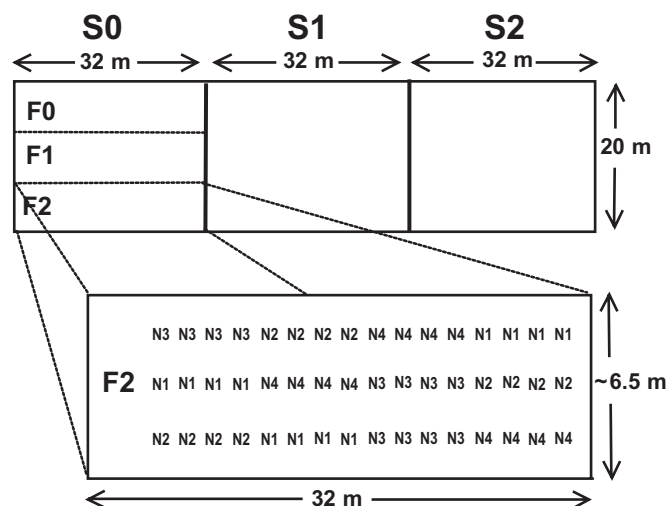
Past research has indicated some direct and indirect impacts of ericaceous shrubs on soil nutritional processes, humus accumulation, conifer root growth, and mycorrhizal infection (Bradley et al. 2000; Damman 1971; Yamasaki et al. 1998; Zhu and Malik 1994). Globally, "growth check" of planted conifers on sites invaded by these species is attributed to competition for nutrients (both direct and indirect), allelopathy, and low soil temperature associated with humus buildup. However, the relative importance of these mechanisms seems to vary according to site quality (Yamasaki et al. 2002).

Given the wide range of characteristics of the sites on which ericaceous species are found in eastern Canada, the related variety of ericaceous communities, and the complex nature of the mechanisms suspected to induce conifer growth-check, local fine-tuning and optimization of management strategies for these sites are clearly needed. We have already conducted various experiments on sites in northwestern Quebec characterized by relatively thin humus (~8 cm) and a *K. angustifolia* – *Vaccinium* spp. complex (Thiffault et al. 2004b, 2005). For these sites, our experiments showed that double-pass scarification is no more beneficial than single-pass treatments in the short term. Similarly, we did not observe any positive interaction between slow-release fertilization and scarification on such sites. The hypothesis that such treatments could alleviate the adverse effects of a *K. angustifolia* – *R. groenlandicum* complex on young conifers on wetter sites with a thick humus layer remains to be verified.

Nutrient loading of seedlings at the nursery is another means to enhance their competitive ability once outplanted in the field (Timmer and Munson 1991). This approach has proved effective on a boreal mixedwood site dominated by American red raspberry, *Rubus idaeus* L. (Malik and Timmer 1996). Given the nutritional nature of the interference on ericad-dominated sites (Thiffault et al. 2004b; Yamasaki et al. 2002), it is hypothesized that planting conifer seedlings with enhanced nutrient reserves on *K. angustifolia* – *R. groenlandicum* sites significantly improves field performance.

The objective of our study was to quantify the main and interaction effects of initial foliar N concentration, scarification, and fertilization on black spruce (*Picea mariana* (Mill.) BSP) seedling survival and 5-year growth on a *K. angustifolia* – *R. groenlandicum*-dominated site in north-eastern Quebec. We also aimed to assess treatment impacts on competing vegetation and selected environmental resources. We hypothesized that higher initial foliar N concentrations and the use of fertilizer favour seedling nutrition and early growth. We also expected scarification to benefit seedling establishment by enhancing availability of critical

Fig. 1. Experimental design, showing an example of one complete block with details of one subplot. Treatments are as follows: S0, no scarification; S1, single-pass scarification; S2, double-pass scarification; F0, no fertilization; F1, fertilization with Forest Paks at time of planting; F2, fertilization with Silva Paks at time of planting. Foliar N concentrations are as follows: N1, 1.43%; N2, 1.89%; N3, 2.33%; N4, 2.76%. Each "N" represents two black spruce (*Picea mariana*) seedlings planted 1 m apart along the row; rows are spaced 2 m apart.



resources, notably soil extractable nutrients, and soil temperature.

Materials and methods

Site description

An experimental plantation was established north of the city of Baie-Comeau, Quebec, Canada, in the black spruce – feathermosses bioclimatic domain designed by Saucier et al. (1998) (49°47'18"N, 69°17'12"W). The regional climate is subpolar subhumid continental, with annual mean temperatures of –2.5–0.0 °C and a growing season of 140 days. Mean annual precipitation is 1300 mm (35%–40% falls as snow). The study site (~5 ha) is located 472 m above sea level and the soil surface deposit is a deep till (>1 m) with a sandy loam texture in the first 15 cm (62% sand; 34% loam; 5% clay). The Humo-Ferric Podzolic soil (Soil Classification Working Group 1998) is moderately well drained and covered with a Mor humus 18 cm deep. Before clear-cutting in 1993, the site supported a *P. mariana* stand more than 120 years old with a site index of 15 m (dominant height at age 50). At cutting, the stand produced 176 m³·ha^{–1} of merchantable wood. Following cutting, a dense cover of ericaceous shrubs (mainly *K. angustifolia* and *R. groenlandicum*) formed and advance regeneration was sparse. These site conditions are typical of *K. angustifolia* heath formation following harvesting in this region.

Experimental design and treatments

In October 1999 we established a split-split-plot experimental design with 10 replicated blocks each 20 m × 100 m. Each block was divided into three main plots 20 m × 32 m with buffer zones 2 m wide between plots. In each block we

Table 1. Seedling characteristics at the time of planting (June 2000).

	Foliar N concentration ^a			
	N1	N2	N3	N4
Dimensions				
Height (cm)	17.8 (2.5)	17.7 (2.5)	18.5 (2.6)	17.9 (2.7)
Ground-level diameter (GLD; mm)	2.9 (0.6)	3.3 (0.6)	3.5 (0.6)	3.5 (0.7)
<i>n</i>	1080	1080	1079	1080
Dry biomass				
Needles (g)	0.98 (0.15)	1.06 (0.18)	1.11 (0.16)	1.06 (0.16)
Twigs (g)	0.13 (0.04)	0.17 (0.12)	0.17 (0.05)	0.16 (0.06)
Stem (g)	0.38 (0.08)	0.47 (0.10)	0.53 (0.10)	0.50 (0.11)
Total aboveground (g)	1.49 (0.24)	1.70 (0.27)	1.81 (0.24)	1.72 (0.29)
<i>n</i>	60	60	60	60

Note: Values are shown as the mean with SD in parentheses.

^aN1, 1.43%; N2, 1.89%; N3, 2.33%; N4, 2.76% (Kjeldahl N, sampled at the nursery in May 2000).

randomly assigned one of the three following mechanical site preparation treatments to each main plot: (1) control (no scarification; S0); (2) single-pass cone scarification with a Wadell scarifier (S1); and (3) double-pass Wadell scarification, the second pass being perpendicular to the first (S2). The site was scarified in October 1999.

In spring 2000 we divided the main plots into three subplots (Fig. 1), each containing 96 seedlings, and randomly assigned to each subplot one of the three following fertilization treatments: (1) no fertilization (F0); (2) fertilization with Forest Paks (Reforestation Technologies International, Salinas, Calif.), a slow-release fertilizer contained in a "teabag", at time of planting (F1); and (3) fertilization with Silva Paks (Reforestation Technologies International), another form of slow-release fertilizer, at time of planting (F2). Each Forest Pak (F1) contains 10 g of fertilizer (15.0% total N; 15.0% P (P₂O₅) available as phosphoric acid; 8.0% K (K₂O) available as soluble potash; 8.20% S; 0.20% Zn; 0.41% Fe; 0.98% Mg; 0.10% Cu; 0.12% B; 0.22% Mn; and 0.09% Mo; this corresponds to 1.5 g elemental N, 0.6546 g elemental P, 0.66408 g elemental K, 0.8 g S, 0.02 g Zn, 0.041 g Fe, 0.098 g Mg, 0.01 g Cu, 0.012 g B, 0.022 g Mn, and 0.009 g Mo). N, P, and K sources are polymer-coated to supply 14.73% slow-release N, 6.64% slow-release P, and 5.49% slow-release potash. Each Silva Pak (F2) contains 10 g of fertilizer (26.3% total N; 12.0% P (P₂O₅) available as phosphoric acid; and 6.0% K (K₂O) available as soluble potash; this corresponds to 2.6 g elemental N, 0.5 g elemental P, and 0.5 g elemental K). N, P, and K sources are polymer-coated to supply 24.9% slow-release N, 5.3% slow-release P, and 5.1% slow-release potash. In the F1 and F2 subplots we buried fertilizer bags 2 cm from the planting holes and 4 cm deep at time of planting.

Seedlings (2 + 0) germinated in 1998 (seed origin 48°20'N, 70°02'W) at the Centre sylvicole Forestville, a private nursery in Quebec, in containers consisting of 45 cells, each 110 cm³. Seedling production was completed in 1999–2000 at the Saint-Modeste provincial nursery. Each subplot (F0, F1, and F2) contained 24 seedlings at each of the following four levels of initial foliar N concentration randomly distributed (at the sub-subplot level): 1.43% (N1), 1.89% (N2), 2.33% (N3), and 2.76% (N4). P concentrations were 0.29%, 0.27%, 0.32%, and 0.32% for the N1, N2, N3, and

N4 treatment, respectively. K concentrations were 0.83%, 0.72%, 0.71%, and 0.73% for the N1, N2, N3, and N4 treatment, respectively. Concentrations are those measured at the nursery, before seedlings were shipped to the field (in May 2000). The N2 treatment corresponds to a standard foliar N concentration for seedling production in Quebec (Ministère des Ressources naturelles et de la Faune du Québec 2005). The desired foliar concentrations were obtained between August and October 1999 with use of the PLANTEC software (Girard et al. 2001), which calculates weekly fertilizer doses by integrating the evolution, over time, of seedling dry biomass, foliar N, P, and K concentrations, and substrate fertility. We planted seedlings in June 2000 at 1 m × 2 m spacing and tagged one seedling in every two (2 m × 2 m) for growth measurements. Extra seedlings were used for destructive sampling. A total of 8640 seedlings were planted.

Seedling measurements

Height (cm) and ground-level diameter (GLD; mm) were measured immediately after planting (Table 1) and in October of each of the first five growing seasons (2000–2004 inclusive). The survival rate was also noted each year. At time of planting (in June 2000) we randomly selected 60 seedlings with each initial foliar N concentration (N1–N4; a total of 240 seedlings) for determining initial aboveground biomass. Seedlings were kept frozen until they were oven-dried at 65 °C for 48 h. We separately weighed needles, twigs, and stems (Table 1). In October of 2000 (first growing season) and 2002 (third growing season) we sampled two seedlings in each scarification × fertilization × initial foliar N concentration combination for every block (a total of 720 seedlings) for determining aboveground biomass. Subsamples (~3.5 g) of dried needles used for biomass measurements in 2000 and 2002 were kept for measuring nutrient concentrations (see below). In 2004 we collected two composite foliar samples (one composite sample included needles from three seedlings) in each sub-subplot in every block. Foliar samples (2000, 2002, and 2004) were ground to 0.5 mm and subjected to Kjeldahl digestion (H₂SO₄/H₂O₂/Se) (Walinga et al. 1995). Kjeldahl N was measured colorimetrically by spectrophotometry (FIA Quickchem, Lachat, Milwaukee, Wis.), and P, K, Ca, and Mg by inductively coupled plasma analysis (ICAP-9000, Thermo Instruments, Franklin, Mass.).

Vegetation assessment

We carried out surveys of vegetation density (the number of individual stems protruding from the soil or humus) in mid-July of the first five growing seasons in one 0.8 m radius (2 m²) permanent vegetation sampling plot within each main plot of every block. Surveys were not exhaustive, but were limited to species recognized as potential competitors in this region (Jobidon 1995).

Soil measurements

From June to October of each study year (2000–2004 inclusive) we measured soil temperature at 10 cm depth in the centre of each main plot (S0, S1, S2) of six experimental blocks using thermistors (model 107 BAM, Campbell Scientific, Inc., Logan, Utah) and a data logger (CR-10, Campbell Scientific, Inc.). Temperature was measured at 15-min intervals and averaged hourly. We assessed soil nutrient concentrations in October of the second (2001), fourth (2003), and fifth (2004) growing seasons by sampling the surface mineral soil (0–10 cm) in the main plots of every experimental block. We took three samples per main plot; each sample was a composite of three subsamples. Soil samples were air-dried down to 5% mass-based humidity at ambient temperature, and ground to pass a 2 mm mesh screen. Mineral N (NH₄-N and NO₃-N) was extracted with a 2 mol/L KCl solution and P, K, Ca, and Mg with Mehlich-3 solution (Sen Tran and Simard 1993). Chemical analyses were performed by spectrophotometry and inductively coupled plasma analysis.

Statistical analyses

Analyses of variance (ANOVA) for a split-split-plot design were conducted to assess treatment effects on survival (years 1, 3, and 5), height (year 5), GLD (year 5), above-ground biomass (year 3), and foliar nutrient concentrations (years 1, 3, and 5). Height, GLD, and biomass were natural-log-transformed prior to the analyses to meet normality and homoscedasticity requirements; for these variables we present back-transformed means and 95% confidence intervals after bias correction (Ung and Végiard 1988). All ANOVAs were performed with the MIXED procedure of the SAS[®] 9.1 system; survival data (binomial) were analysed with the macro procedure GLIMMIX (Littell et al. 1996). Given the high occurrence of experimental units without any mortality (19 out of 36), survival modelling was only possible by attributing a “dead” value to 1 sample seedling (out of 120) from each scarification × fertilization × initial N concentration combination for which it was the case. For all these variables we constructed a priori contrasts to answer specific questions: (i) is there an advantage to scarification, regardless of the method (S1 + S2 vs. S0)? (ii) if we scarify, is double-pass scarification better than single-pass scarification (S2 vs. S1)? (iii) is there an advantage to fertilization, regardless of the method (F1 + F2 vs. F0)? (iv) if we fertilize, is Forest Pak better than Silva Pak (F2 vs. F1)? (v) is the standard foliar N concentration better than a lower concentration (N2 vs. N1)? (vi) is there an advantage to using seedlings above the standard foliar N concentration (N3 + N4 vs. N2)? and (vii) if we use seedlings above the standard, is 2.76% foliar N better than 2.33% foliar N (N4 vs. N3)? An α value of 0.05 or less was considered significant.

Seedling height profiles over 5 years were compared among treatments using ANOVA for repeated measures. We performed these analyses with the REPEATED statement of the MIXED procedure (Littell et al. 1996) on natural-log-transformed data. A first-order autoregressive covariance structure (AR(1)) was selected. Given the relatively complicated covariance structure and the reasonably balanced design, we used the Kenward–Roger method for calculating denominator degrees of freedom (Schaalje et al. 2001). Treatment comparisons by contrasts were performed with Helmert transformations, which compare a level of a repeated measure with the mean of subsequent levels (SAS Institute Inc. 2004).

We used principal component analyses (PCA) to study relationships between the different species and the scarification treatments for each sampling year. The FACTOR procedure of SAS[®] was used to extract principal components. As stem density varies greatly from one species to another, PCAs were performed using correlation matrix, which forces all variables to have equal variances. Scores of the scarification treatments on axes 1 and 2 of each PCA were compared by means of ANOVAs after being assessed for agreement with standard parametric statistical assumptions. We used Fisher’s protected LSD test to separate treatment means when ANOVAs showed significant effects ($p < 0.05$). Given the lack of new meaningful information brought with the analyses of data from the second, third, and fourth seasons, we only report results using the first- and fifth-year data.

The effects of scarification and its intensity on extractable soil nutrient concentrations in 2001 and 2003 were evaluated with ANOVAs for completely randomized block design. Data were natural-log-transformed to comply with ANOVA prerequisites. We compared soil temperature regimes against scarification treatments during the first, third, and fifth growing seasons using ANOVA for repeated measures; a spatial power covariance matrix was used (Littell et al. 1996).

Results

Seedling dimensions and survival rate

Scarification and fertilization at planting significantly interacted to influence seedling height and diameter at age 5 and aboveground biomass after three growing seasons (Table 2, Figs. 2 and 3). In all cases, double-pass scarification (S2) resulted in taller and bigger seedlings than single-pass scarification (S1) (Table 3). Independently of scarification or initial N concentration, seedlings fertilized with Forest Paks (F1) had greater aboveground biomass than seedlings fertilized with Silva Paks (F2) (Table 3). The positive effect of scarification (either S1 or S2) on GLD and biomass was enhanced when seedlings were fertilized (with either F1 or F2) (Table 3, Fig. 3). Although significant in some cases (e.g., N2 vs. N1, $p \leq 0.002$ for GLD and biomass), the effect of initial N concentration was weak compared with other treatment effects (Figs. 2 and 4).

Seedling mortality was scant after one growing season (October 2000), being less than 1% in any treatment combination. After three growing seasons (October 2002) fertilization had a significant adverse effect on seedling survival ($p <$

Table 2. Summary of ANOVA results (*p* values) for some seedling-response variables.

Effect (fixed)	dfn	dfd	Height (fifth year)	GLD (fifth year)	Biomass (third year)	Foliar nutrient concentration (fifth year)				
						N	P	K	Ca	Mg
Scarification (S)	2	27	<0.001	<0.001	<0.001	0.007	<0.001	0.045	0.015	0.057
Fertilization (F)	2	55	<0.001	<0.001	<0.001	0.006	0.010	0.514	0.015	<0.001
S × F	4	55	0.030	0.018	0.007	0.253	<0.001	0.063	0.120	0.223
Initial foliar N concn. (N)	3	242	<0.001	<0.001	<0.001	0.510	0.080	0.127	0.393	0.020
S × N	6	242	0.895	0.736	0.999	0.429	0.191	0.014	0.619	0.948
F × N	6	241	<0.001	<0.001	0.001	0.563	0.026	0.011	0.402	0.670
S × F × N	12	241	0.064	0.349	0.476	0.497	0.224	0.603	0.950	0.325

Note: Analyses of height, GLD, and biomass were performed on natural-log-transformed data (dfn is numerator degrees of freedom for all variables; dfd is denominator degrees of freedom for height, calculated using Satterthwaite’s formula (Littell et al. 1996); dfd for other variables varies from 18 to 297). Given the significant interactions, values in boldface type are for interpretation.

Fig. 2. Height profiles over 5 years for scarification and fertilization combinations (A) and initial foliar N concentration and fertilization combinations (B). Analyses performed on natural-log-transformed data, back-transformed means, are presented (Ung and Végiard 1988). For details of treatments and foliar N concentrations see Fig. 1.

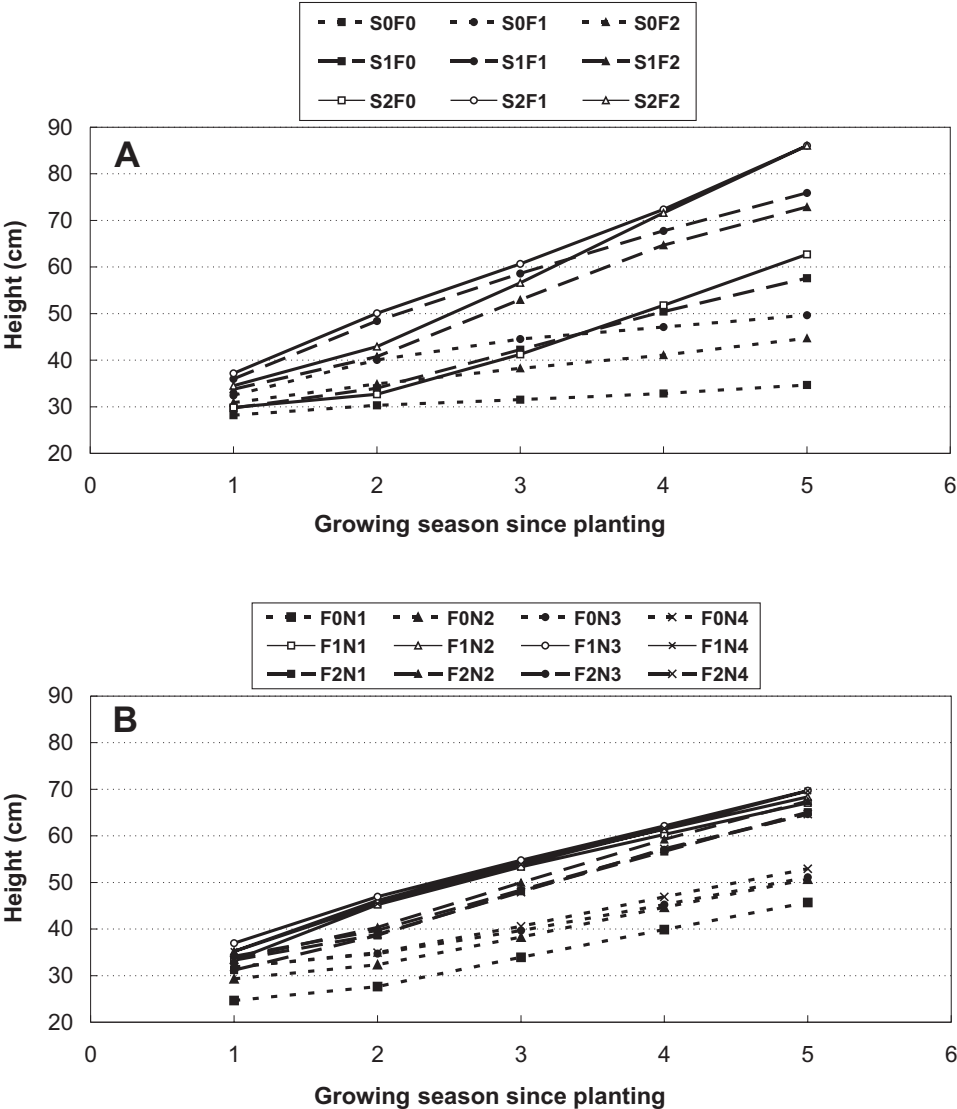


Fig. 3. Scarification and fertilization effects on seedling dimensions after three (biomass) or five (ground-level diameter, GLD) growing seasons. Analyses were performed on natural-log-transformed data. Back-transformed means and 95% confidence intervals are shown (after Bernier-Cardou and Bigras 2001). For details of treatments see Fig. 1.

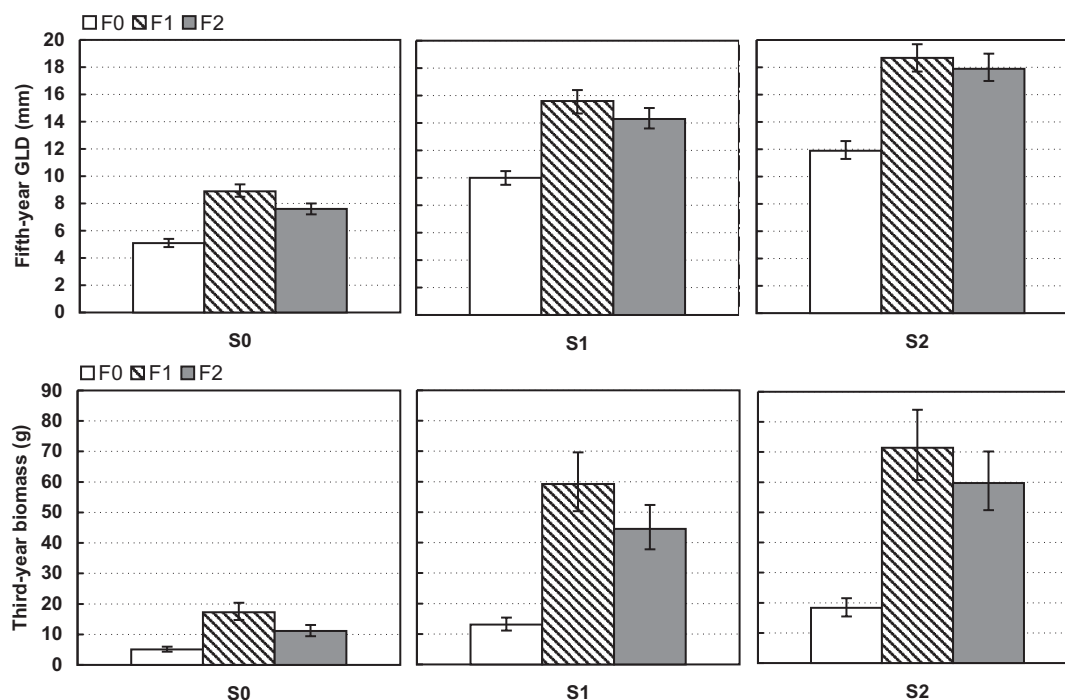
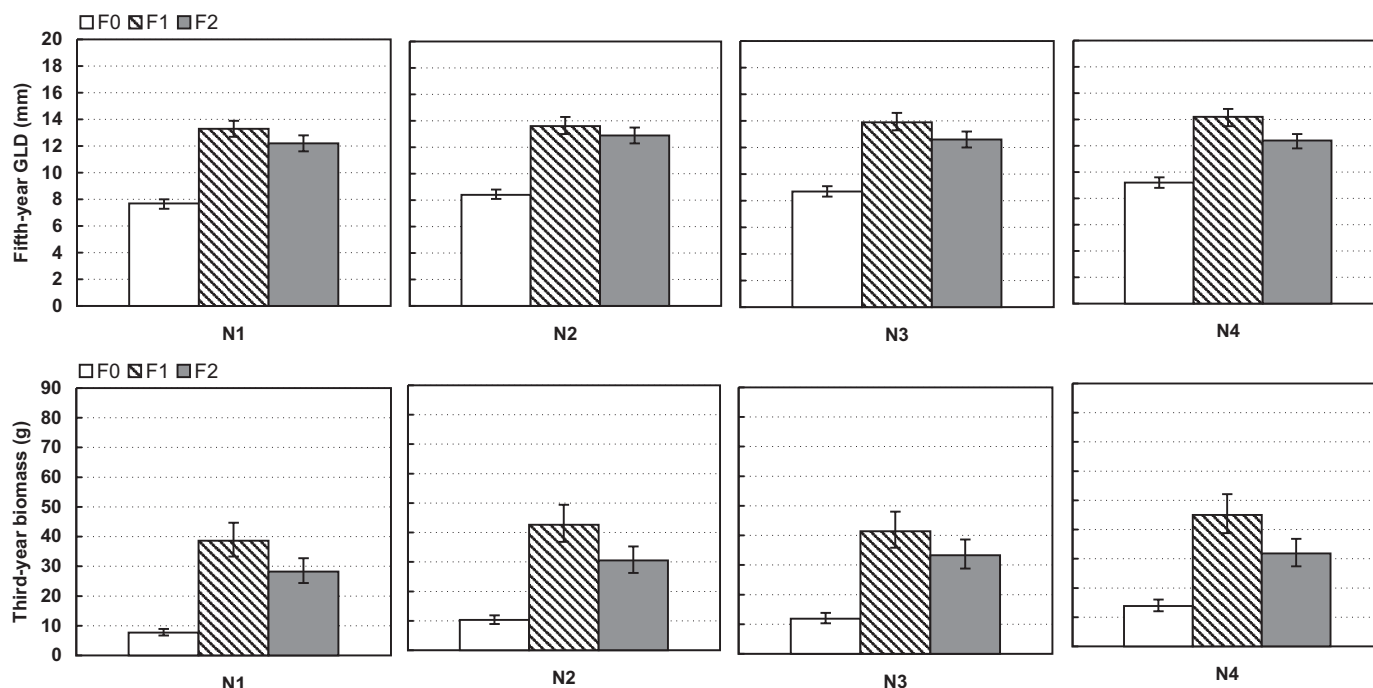


Table 3. Results of planned comparisons for some seedling-response variables.

Contrast	Fifth-year height		Fifth-year GLD		Third-year biomass	
	df	<i>p</i> > <i>F</i>	df	<i>p</i> > <i>F</i>	df	<i>p</i> > <i>F</i>
(1) S1 + S2 vs. S0	27.0	<0.001	18.1	<0.001	18	<0.001
(2) S2 vs. S1	27.0	0.002	18.0	<0.001	18	0.002
(3) F0 vs. F1 + F2	53.9	<0.001	54.2	<0.001	54	<0.001
(4) F1 vs. F2	56.3	0.002	56.6	<0.001	54	<0.001
(1) × (3)	54.0	0.458	54.4	0.039	54	0.002
(1) × (4)	56.5	0.013	56.8	0.011	54	0.064
(2) × (3)	53.8	0.071	54.1	0.455	54	0.417
(2) × (4)	56.0	0.307	56.4	0.333	54	0.428
(5) N2 vs. N1	239.0	<0.001	241.0	<0.001	243	0.002
(6) N3 + N4 vs. N2	238.0	0.988	240.0	0.122	243	0.015
(7) N4 vs. N3	246.0	0.404	248.0	0.209	243	0.207
(1) × (5)	243.0	0.811	245.0	0.391	243	0.904
(1) × (6)	240.0	0.785	242.0	0.567	243	0.900
(1) × (7)	246.0	0.556	248.0	0.448	243	0.849
(2) × (5)	235.0	0.227	237.0	0.190	243	0.663
(2) × (6)	237.0	0.287	239.0	0.190	243	0.765
(2) × (7)	245.0	0.716	248.0	0.876	243	0.861
(3) × (5)	231.0	0.003	232.0	0.057	243	0.052
(3) × (6)	231.0	0.064	233.0	0.011	243	0.034
(3) × (7)	234.0	0.137	236.0	0.049	243	0.197
(4) × (5)	247.0	0.529	249.0	0.369	243	0.847
(4) × (6)	246.0	0.010	248.0	0.026	243	0.601
(4) × (7)	257.0	0.910	260.0	0.274	243	0.264

Note: Analyses were performed on natural-log-transformed data. Degrees of freedom were calculated using Satterthwaite's formula (Littell et al. 1996). Treatments are as follows: S0, no scarification; S1, single-pass scarification; S2, double-pass scarification; F0, no fertilization; F1, Forest Paks fertilization; F2, Silva Paks fertilization. Foliar N concentrations are as follows: N1, 1.43%; N2, 1.89%; N3, 2.33%; N4, 2.76%. Given the significant interactions, only values in boldface type are for interpretation.

Fig. 4. Initial foliar N concentrations and fertilization effects on seedling dimensions after three (biomass) or five (GLD) growing seasons. Analyses were performed on natural-log-transformed data. Back-transformed means and 95% confidence intervals are shown (after Bernier-Cardou and Bigras 2001). For details of treatments and foliar N concentrations see Fig. 1.



0.001). Forest Pak fertilization (F1) was associated with 3% mortality, whereas seedlings fertilized with Silva Pak (F2) exhibited 13%, a difference that was highly significant ($p < 0.001$). Ninety-nine percent of the unfertilized seedlings remained alive until October 2002, a proportion that was significantly higher than for both types of fertilized seedlings ($p \leq 0.012$). We observed the same pattern after five growing seasons (October 2004); mortality was 1%, 3%, and 14% for the F0, F1, and F2 treatment, respectively. Scarification and initial foliar N concentration treatments did not induce noticeable seedling mortality after five growing seasons.

Height profiles

Seedling heights were greater in the scarified plots (S1 and S2) than in the unscarified ones (S0); this difference increased with time (Fig. 2). However, scarification and fertilization interacted through time and influenced seedling height progression over the 5 years of our study, in two ways (Table 4). First, the effect of scarification on seedling height increased more for fertilized (F1 and F2) than for unfertilized (F0) seedlings (Fig. 2). Second, the beneficial effect of soil scarification (S1 and S2) over no soil treatment (S0) was enhanced for F1 seedlings compared with F2 seedlings until year 3. After that (from year 3 to year 5), F2 seedlings responded to scarification slightly better than F1 seedlings. For a given fertilization treatment, S1 and S2 height profiles were statistically parallel from year 2 to year 5. Fertilization also interacted with initial foliar N concentration and time (Table 4). The positive effect of fertilization on height was maximized at the end of the second growing season for the seedlings with the lowest initial N concentration (N1) (Fig. 2). Afterward, the effect remained constant from year 3 to year 5. Height profiles of the other fertilization $\times$ initial N

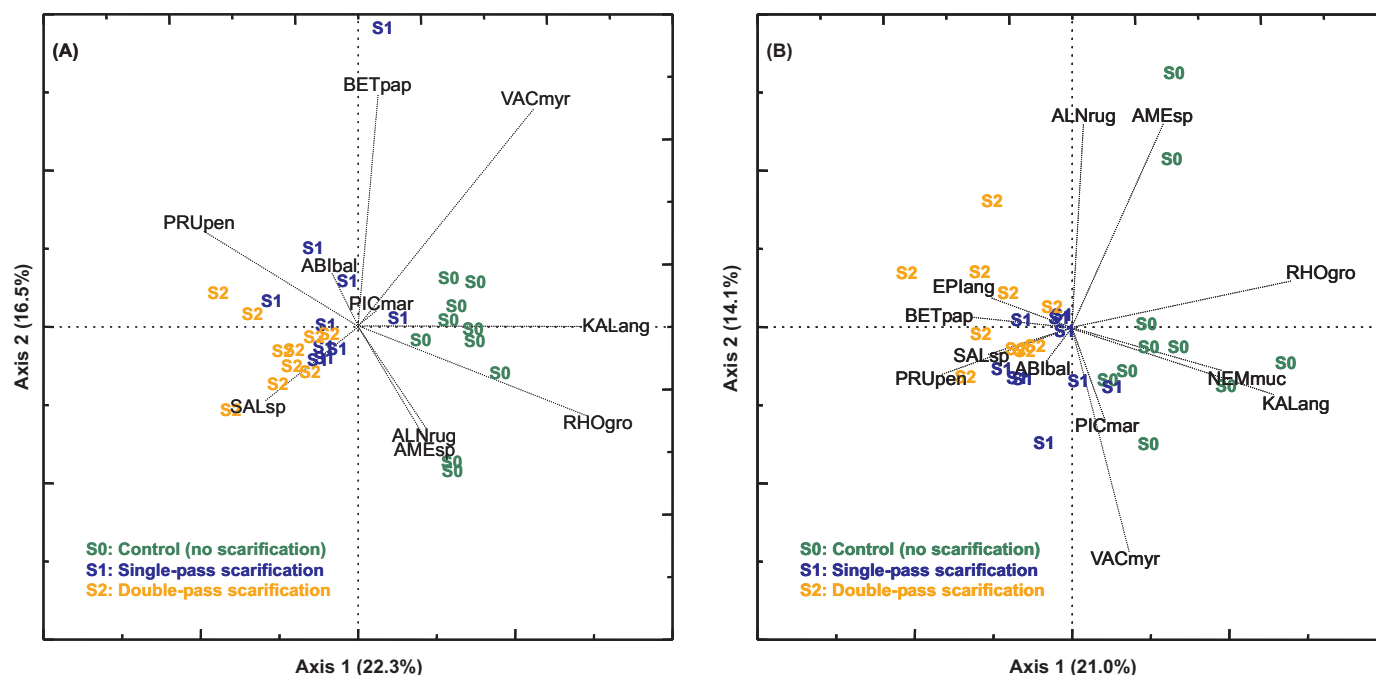
Table 4. Results of ANOVA for repeated measurements on seedling height over 5 years.

Effect (fixed)	dfn	dfd	F	$p > F$
Scarification (S)	2	27	138.5	<0.001
Fertilization (F)	2	307	574.2	<0.001
S $\times$ F	4	231	5.0	<0.001
Initial foliar N concn. (N)	3	307	25.8	<0.001
S $\times$ N	6	245	0.6	0.729
F $\times$ N	6	307	11.5	<0.001
S $\times$ F $\times$ N	12	265	1.2	0.307
Time (T)	4	1188	5363.8	<0.001
S $\times$ T	8	1033	385.9	<0.001
F $\times$ T	8	1198	117.8	<0.001
S $\times$ F $\times$ T	16	1120	11.0	<0.001
N $\times$ T	12	1201	9.3	<0.001
S $\times$ N $\times$ T	24	1155	1.2	0.248
F $\times$ N $\times$ T	24	1205	1.7	0.019
S $\times$ F $\times$ N $\times$ T	48	1188	0.8	0.865

Note: dfn is numerator degrees of freedom; dfd is denominator degrees of freedom, calculated using the Kenward–Roger formula (Schaalje et al. 2001). Given the significant interactions, values in boldface type are for interpretation.

concentration combinations were statistically parallel; we were thus able to examine the initial N concentration $\times$ time interaction. Firstly, the analysis of contrasts for this interaction indicated that the height difference between N1 and N2 seedlings diminished from year 1 to year 2 and then remained constant from year 3 onward. Secondly, it showed that the better height growth of N3 and N4 seedlings than of N2 seedlings during years 1 and 2 disappeared after two

Fig. 5. Principal component analyses of competing species by scarification treatment in the first growing season (2000) (A) and the fifth growing season (2004) (B). Species are designated as follows: ABIBal, *Abies balsamea*; ALNrug, *Alnus rugosa*; AMEsp, *Amelanchier* sp.; BETpap, *Betula papyrifera*; EPIang, *Epilobium angustifolium*; KALang, *Kalmia angustifolia*; NEMmuc, *Nemopanthus mucronatus*; PICmar, *Picea mariana*; PRUpen, *Prunus pensylvanica*; RHOGro, *Rhododendron groenlandicum*; SALsp, *Salix* sp.; VACmyr, *Vaccinium myrtilloides*.



growing seasons; the height profiles were parallel and superposed during the 3 following years.

Foliar nutrient concentrations

The treatments did not interact to influence seedling foliar N, Ca, and Mg concentrations after five growing seasons (Table 2). Planned comparisons showed that S1 or S2 scarification (mean fifth-year foliar N concentration 0.87%) had a positive impact compared with S0 (fifth-year foliar N concentration 0.82%) ($p = 0.019$), whereas it significantly reduced the needle Ca concentration (by 0.03%; $p = 0.016$) and had no significant effect on Mg concentration ($p = 0.057$). The superiority of S2 over S1 (0.06%) was significant for N concentration at year 5 ($p = 0.020$) but not for Ca concentration ($p = 0.074$). Fertilization at time of planting decreased fifth-year foliar N concentration by 5% compared with F0 ($p = 0.002$), and slightly reduced Ca and Mg concentrations ($p \leq 0.010$). The F1 treatment, in which micro-nutrients were included in the formulation, significantly enhanced foliar Mn ($p < 0.001$) and Zn ($p < 0.001$) concentrations compared with the F2 treatment after one growing season. The effect was no longer apparent in 2002 (year 3) for either nutrient ($p \geq 0.250$). We measured higher foliar N concentrations in F2 seedlings than in F1 seedlings in 2000 (year 1) and 2002 (an average increase of 0.3%; $p < 0.001$), but only in unscarified plots in the latter case. For N, Ca, and Mg concentrations we detected no significant difference between F1 and F2 ($p \geq 0.176$) in 2004 (year 5), or between most of the initial foliar N concentration treatments (Table 2). Fertilization, either F1 or F2, significantly reduced the foliar P concentration in N1 seedlings compared with F0

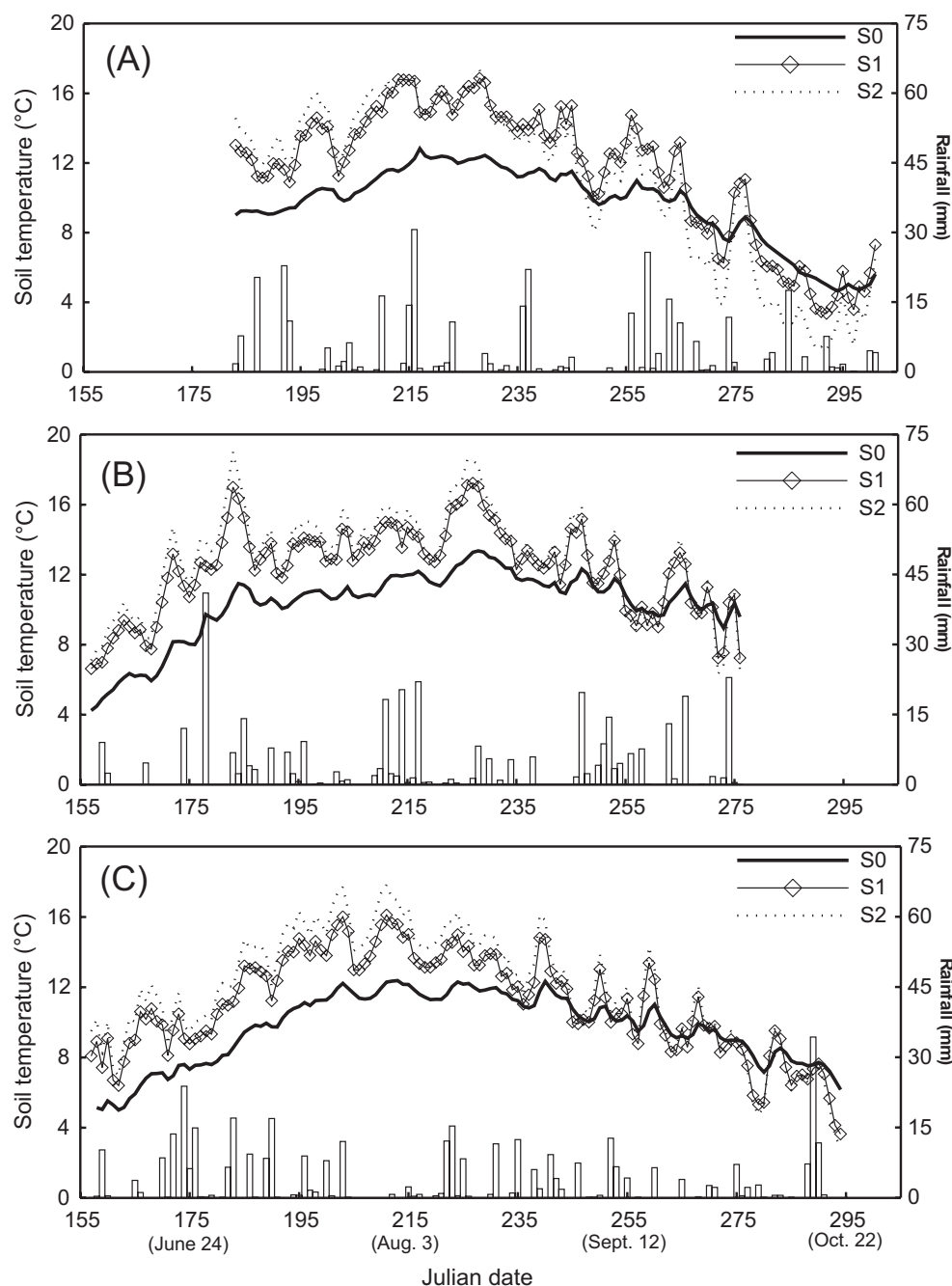
($p < 0.001$) in 2004, but had no effect in the other initial N concentration treatments ($p \geq 0.352$). Although significant during the fifth season, the interaction between fertilization and initial N concentration for foliar K concentration (Table 2) revealed no significant difference in the contrasts ($p \geq 0.057$). Analysis of fifth-year foliar P concentrations (Table 2) revealed a significant positive effect of scarification for all fertilization levels, combined with a detrimental impact of fertilization in S0 plots.

Vegetation

In 2000 (year 1), the first axis of the PCA (eigenvalue = 2.23) represented opposition between ericaceous plants (*K. angustifolia*, *R. groenlandicum*, and *V. myrtilloides*) and two northern deciduous tree species (pin cherry (*Prunus pensylvanica* L.f.) and willow (*Salix* sp.)) (Fig. 5A). *Betula papyrifera* and *V. myrtilloides* were highly associated with the second axis (eigenvalue = 1.65). As in 2000, the first axis in 2004 (year 5; eigenvalue = 2.52) represented opposition between ericaceous plants (*K. angustifolia* and *R. groenlandicum*) and *P. pensylvanica*. *Betula papyrifera* and *Salix* sp. were also negatively associated with this axis (Fig. 5B). The second axis (eigenvalue = 1.70) was clearly associated with *Amelanchier* sp. and *Alnus rugosa* on one hand and *V. myrtilloides* on the other.

In 2000 and 2004, scarification significantly influenced the composition of the vegetation sampling plots described by the first principal component of the PCA ($p < 0.001$). During the first and fifth growing seasons, control plots (S0) were strongly associated with positive values on axis 1, whereas the S1 and S2 plots were associated with negative

Fig. 6. Soil temperature regimes and daily rainfall in 2000 (A), 2002 (B), and 2004 (C). Treatments are as follows: S0, no scarification; S1, single-pass scarification; S2, double-pass scarification.



values. Scores for the S2 treatment were significantly more negative than scores for the S1 treatment in 2000 and 2004 ($p \leq 0.011$). Scores on axis 2 were not significantly influenced by scarification ($p \geq 0.122$).

Soil characteristics

Scarification did not significantly affect soil extractable N-NH₄, K, or Mg in either 2001 (year 2) or 2003 (year 4; $p \geq 0.101$). In 2001, P concentrations were 30% lower in scarified plots than in control plots ($p = 0.001$); this effect was no longer significant in 2003. Scarification, either sin-

gle- or double-pass, significantly enhanced soil extractable Ca concentrations in 2003 ($p \leq 0.001$). For all treatments and in all years, extractable N-NO₃ concentrations were below detection limits. Soil temperature profiles of the scarification treatments throughout the first, third, and fifth seasons were not parallel (scarification $\times$ time, $p \leq 0.001$). From early June to early September of each year, the soil temperature was systematically higher in the S1 and S2 plots than in the S0 plots (Fig. 6). Between Julian Days 157 and 250, a mean difference of 3.2 °C was noted in the first year, 3.3 °C in the third year, and 2.8 °C in the fifth year. The differences be-

tween the treatments gradually disappeared in September and October of each season.

Discussion

Scarification had a strong positive impact on seedling growth. The growth differences between the seedlings planted in scarified and unscarified plots appeared early in our study and increased over time. As a result, after 5 years, seedlings planted in scarified soil were almost twice the height and diameter of those planted in unscarified plots. The positive impact of scarification on seedling biomass was even greater, ranging from a 208% gain for F0 seedlings to a 369% gain for F2 seedlings. Such positive responses in seedling growth to mechanical soil preparation have been reported for northeastern sites in Canada invaded by ericaceous shrubs. Prévost and Dumais (2003) demonstrated the advantage of scarification for growth of *P. mariana* on boreal sites. A similar advantage of scarification was observed by Thiffault et al. (2005) 3 years after planting *P. mariana* and jack pine (*Pinus banksiana* Lamb.) on a *K. angustifolia* – *Vaccinium* spp. site in northwestern Quebec. On the other hand, the height-growth trajectories we observed in unscarified areas are typical of stunted seedlings (Mallik 2001; Titus et al. 1995), which will likely remain weak for several decades (Jobidon 1995; Mallik 2003).

Scarification was associated with the emergence of species that were not observed in control plots; the treatments favoured the emergence of intolerant, early-successional species and significantly reduced the abundance of ericaceous shrubs. The appearance of *P. pensylvanica*, a typical “seed bank” species (Marks 1974), was clearly stimulated by scarification. Based on results reported by Auchmoody (1979), we hypothesize that increased nitrification following soil disturbances (Lavoie and Bradley 2003) stimulated germination of buried seeds. On the other hand, seeds of *Salix* sp. are wind-dispersed and are known to germinate within 24 h when they land on well-lit, exposed mineral soils (Louter et al. 1993), as are found in scarified plots.

We did not measure any significant effect of scarification on soil extractable N after two or four growing seasons. Nevertheless, seedlings planted in scarified plots exhibit a higher foliar N concentration than those planted in control plots, even with the dilution effect of increased biomass (Timmer 1991). Similarly, the small but significant negative impact of site preparation on soil extractable P did not result in lower foliar P concentrations. Although all scarification × fertilization combinations led to P concentrations within the critical range reported by Morrison (1974) (0.14%–0.18%), seedlings in scarified plots had enhanced foliar P concentrations compared with seedlings in S0 plots. These results indicate that scarification somehow favoured nutrient absorption, most probably because of enhanced root growth (Nordborg and Nilsson 2003), stimulated by a higher soil temperature (Kaspar and Bland 1992). The sum of soil degree-days above 5 °C was 43% higher in 2002 and 40% higher in 2004 in scarified plots than in control plots ($p \leq 0.003$). Such increases are similar to those reviewed by Örlander et al. (1990) for sites in Fennoscandia. However, our study does not take into account all the potential direct and indirect impacts of the treatments, such as on the or-

ganic nutrient pools (Hamel et al. 2004; Näsholm et al. 1998) or the N₂-fixing potential of the feather moss layer (Houle et al. 2006).

Seedlings planted in plots subjected to double-pass scarification (S2) grew taller and bigger than seedlings planted in S1 plots. In terms of height, the advantage of S2 treatment over S1 was expressed during the first two growing seasons. Afterward, the difference between the two treatments remained constant. As available soil nutrients, soil temperature regime, percent organic matter, cation-exchange capacity, and sum of soil degree-days (data not shown) were all statistically similar for the two treatments, we believe this difference to be the result of a reduction in the direct influence of the ericaceous shrubs on the seedlings during the first two seasons. Both scarification treatments significantly modified the vegetation communities compared with the control, but the influence of S2 was stronger than that of S1. Double-pass scarification has also proved more effective than single-pass scarification in reducing *K. angustifolia* cover on a *K. angustifolia* – *Vaccinium* spp. site during the first growing season (Thiffault et al. 2005). In this study, the second pass of the scarifier also significantly increased the distance between the planted seedlings and the nearest *K. angustifolia* stem (Thiffault et al. 2005). The differences were no longer apparent after three growing seasons. Although we did not measure the ericaceous cover around each seedling in the present experiment, we suspect the scarification effects to have been similar. Yamasaki et al. (1998) have shown the distance-dependant relationship between planted seedlings and ericaceous shrubs. It thus appears that the larger “ericaceous-shrub-free” area around the seedlings created by the S2 treatment than by S1 was sufficient to explain the better height growth observed during the first 2 years.

Fertilization at the time of planting was beneficial to seedling growth, although some nutrient concentrations were slightly reduced (most probably following dilution). At the same time, scarification effects on seedling growth and fifth-year dimensions were enhanced when combined with fertilization. Fertilizer release from the polymer coating is partly governed by temperature (N. Anderson, Reforestation Technologies International, 1341 Dayton Street, Salinas, Calif., personal communication). We thus suggest that scarification and fertilization interacted as follows: as the warmer soil in the scarified plots promoted seedling root growth (Kaspar and Bland 1992), it also had a beneficial effect on fertilizer release, in turn promoting seedling nutrition and seedling growth. Brand (1991) also reported positive interactions between soil modification and addition of slow-release fertilizers, confirming that efficient use of a resource is a function not only of its availability but also of other environmental factors that affect a seedling's ability to use that resource. The 15N–15P–8K plus micronutrients slow-release fertilizer (F1) proved slightly more effective than the 26N–12P–6K formulation (F2) in terms of seedling growth (a fifth-year gain in height ranging from 11% in S0 plots to 33% in S1 and S2 plots), even if the foliar N concentration was initially better with the higher fertilization rate (until year 3). When they are limiting, responses to the addition of a very small amount of micronutrients can be striking (Binkley 1986); we measured enhanced Mn and Zn concentrations in F1 seed-

lings compared with F2 seedlings after one growing season. Fan et al. (2002) also reported increases in seedling growth with the use of a micronutrient fertilizer compared with macronutrient amendment only.

Our results concur with those from previous studies (Thiffault et al. 2004a, 2005) and further indicate that ericaceous shrub – conifer interactions rarely cause seedling mortality in the short term; the lowest survival rates were observed not in control plots (no scarification, no fertilization) but in plots where seedlings were fertilized. The F2 treatment affected seedling survival rapidly, that is within the first three growing seasons. Although in a lesser proportion than for F2, the F1 treatment also induced mortality during the first three seasons. Simpson and Vyse (1995), as well as Rose and Ketchum (2003), also reported an increase in mortality due to fertilization 7 and 4 years following planting, respectively. Osmotic shock induced by the fertilizer is probably one cause of the mortality that we observed (van Den Driessche 1991). Jacobs et al. (2004) confirmed that high fertilizer salt concentrations in the Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco) seedling rhizosphere induce predawn xylem water potentials lower than those measured in unfertilized seedlings; water stress is the main cause of mortality during the first few years following planting (Burdett 1990; Margolis and Brand 1990). It is also possible that the high nutrient availability associated with fertilization at the time of planting promoted seedling growth beyond the usual period of frost hardening, thus exposing the newly planted seedlings to lethal frost damage. This interpretation is compatible with the fact that mortality was much higher in the F2 treatment (26% N) than in the F1 treatment (15% N).

Nursery × site preparation interaction trials are of particular interest in developing innovative reforestation strategies (South et al. 2001). Based on previous studies (e.g., Boivin et al. 2004), we expected seedling growth in the field to be significantly enhanced by the larger N reserves built up following intensive nutrient loading at the nursery. However, improvements in seedling performance related to increased fertility due to nutrition practices are expected to be short-lived (Grossnickle 2000). Our results support this hypothesis; the N3 and N4 seedlings showed a slight advantage over the seedlings with the standard foliar N concentration during the first 2 years, but their growth curves joined afterward. After five growing seasons, all seedling types had similar foliar N concentrations (0.85%). In comparison, natural seedlings collected in 2001 from control plots throughout the experimental site ($n = 16$) had foliar nutrient concentrations of 1.08% N, 0.18% P, and 0.59% K. As suggested by Salifu and Timmer (2001), the short-term growth response to nutrient loading can be expected to be enhanced on fertile sites compared with poor sites. Nevertheless, our study confirms the importance of foliar N concentration as a factor in seedling quality (Burdett 1990); seedlings below Quebec's standard foliar N concentration (N1 in our study) incur initial growth losses that are carried through years and possibly never recovered.

Conclusion

We hypothesized that scarification benefits seedling estab-

lishment by enhancing the availability of critical environmental resources. In accordance with our hypothesis, the results illustrate a major microsite influence on early seedling growth, and how soil temperature and the proximity of ericaceous shrubs must be managed for the other treatments to be effective. Successful regeneration of many sites with ericaceous shrubs is conditional on periodic removal of the humus layer by wildfires (Mallik 2003). Scarification of these sites can be considered a way to emulate natural processes, as it provides the conditions necessary for resetting secondary succession where ecosystem retrogression is observed following *K. angustifolia* – *R. groenlandicum* invasion. Based on profile analyses, we demonstrated how typical exponential seedling growth is only expressed when the soil is scarified. Nutritional issues, addressed in our hypotheses relating to fertilization and initial foliar N concentration, were clearly revealed once the soil was scarified. Caution should be used in extrapolating these results to other ericaceous-shrub-dominated sites, as interactions may occur between treatment effects and site characteristics. However, ongoing studies on other *K. angustifolia* sites should yield results on a broader scale.

Management implications

We demonstrated that soil scarification is a necessary operation for successful growth of seedlings planted on a *K. angustifolia* – *R. groenlandicum* site. Moreover, our results concur with those reported by Prévost and Dumais (2003) and by Thiffault et al. (2004a), who pointed out that planting to increase natural regeneration stocking is not recommended without proper soil scarification. A small additional gain in height over 5 years can be obtained from a double-pass scarification treatment. However, because of its high cost and small benefit, such a treatment cannot be recommended without a demonstration of longer term benefits.

We identified distinctive vegetation communities in control and scarified plots, the latter being closely associated with early-succession species that were absent from the former. Increased species diversity is to be expected when soil preparation reduces but does not eliminate previously abundant late-successional species (such as *K. angustifolia*) and permits the appearance of early-successional species (such as *P. pensylvanica*) (Swindel et al. 1983, 1984). Mechanical site preparation created regeneration niches for disturbance-dependent indigenous plants without accelerating non-native species invasion, an important characteristic of silviculture based on natural dynamics (Haeussler et al. 2002). A shift in species dominance following scarification is expected to influence the quality of future litter inputs and further modify soil processes (Bloom and Mallik 2004; Bradley et al. 1997).

When combined with soil scarification, spot fertilization at the time of planting provided an additional growth advantage. Based on survival rates, the Forest Pak product, which contains micronutrients and has less N, appears preferable to the Silva Pak for these sites. However, the economic advantage of such treatment on these sites needs to be demonstrated, based on longer term results. Finally, we do not recommend producing seedlings with a higher N concentration than the ones currently under production, the growth gains being short-lived.

Acknowledgements

We sincerely thank Daniel Boudreault, Francis Cadoret, Jacques Carignan, Steve Lemay, Réjean Poliquin, and several summer students for skilful technical work. We are indebted to the staff of the Laboratoire de chimie organique et inorganique of the Direction de la recherche forestière (Ministère des Ressources naturelles et de la Faune du Québec, MRNF), who performed all the chemical analyses. Our thanks also go to Lise Charette and Josianne DeBlois (MRNF) for their helpful statistical advice, to Jacques Duval (MRNF), Charles Warren, Charles Gauthier, Jean-Pierre Michaud, and Jean-François Gauthier (Abitibi Consolidated Inc.) for their kind collaboration in fieldwork, and to Jean-Yves Guay (MRNF), who supervised seedling production at the nursery. Constructive comments from two anonymous reviewers and the guest associate editor are gratefully acknowledged.

References

- Anonymous. 1972. The black spruce regeneration problem in Ontario. *Gt. Lakes For. Res. Cent. Res. Newsl.* 2.
- Auchmoody, L.R. 1979. Nitrogen fertilization stimulates germination of dormant pin cherry seed. *Can. J. For. Res.* 9: 514–516.
- Bernier-Cardou, M., and Bigras, F.J. 2001. The analysis of cold hardiness experiments. In *Conifer cold hardiness*. Kluwer Academic, Boston, Mass. pp. 403–435.
- Binkley, D. 1986. *Forest nutrition management*. Wiley-Interscience, New York.
- Bloom, R.G., and Mallik, A.U. 2004. Indirect effects of black spruce *Picea mariana* cover on community structure and function in sheep laurel (*Kalmia angustifolia*) dominated heath of eastern Canada. *Plant Soil*, 265: 279–293.
- Boivin, J.R., Salifu, K.F., and Timmer, V.R. 2004. Late-season fertilization of *Picea mariana* seedlings: intensive loading and outplanting response on greenhouse bioassays. *Ann. For. Sci.* 61: 737–745.
- Bradley, R.L., Titus, B.D., and Fyles, J.W. 1997. Nitrogen acquisition and competitive ability of *Kalmia angustifolia* L., paper birch (*Betula papyrifera* Marsh.) and black spruce (*Picea mariana* (Mill.) B.S.P.) seedlings grown on different humus forms. *Plant Soil*, 195: 209–220.
- Bradley, R.L., Titus, B.D., and Preston, C.P. 2000. Change to mineral N cycling and microbial communities in black spruce humus after additions of $(\text{NH}_4)_2\text{SO}_4$ and condensed tannins extracted from *Kalmia angustifolia* and balsam fir. *Soil Biol. Biochem.* 32: 1227–1240.
- Brand, D.G. 1991. The establishment of boreal and sub-boreal conifer plantations: an integrated analysis of environmental conditions and seedling growth. *For. Sci.* 37: 68–100.
- Burdett, A.N. 1990. Physiological processes in plantation establishment and the development of specifications for forest planting stock. *Can. J. For. Res.* 20: 415–427.
- Damman, A.W.H. 1971. Effect of vegetation changes on the fertility of a Newfoundland forest site. *Ecol. Monogr.* 41: 253–270.
- Fan, Z., Moore, J.A., Shafii, B., and Osborne, H.L. 2002. Three-year response of Ponderosa pine seedlings to controlled-release fertilizer applied at planting. *West. J. Appl. For.* 17: 154–164.
- Girard, D., Gagnon, J., and Langlois, C.-G. 2001. PLANTEC: a software to manage the fertilization of seedlings in forest tree nurseries. Note de recherche forestière 111, Direction de la recherche forestière, Ministère des Ressources naturelles, Québec, Que.
- Grondin, P., and Cimon, A. 2003. Les enjeux de biodiversité relatifs à la composition forestière. Direction de la recherche forestière et Direction de l'environnement forestier, Ministère des Ressources naturelles, de la Faune et des Parcs, Québec, Que.
- Grossnickle, S.C. 2000. *Ecophysiology of northern spruce species: the performance of planted seedlings*. NRC Research Press, Ottawa, Ont.
- Haeussler, S., Bedford, L., Leduc, A., Bergeron, Y., and Kranabetter, J.M. 2002. Silvicultural disturbance severity and plant communities of the southern Canadian boreal forest. *Silva Fenn.* 36: 307–327.
- Hamel, B., Bélanger, N., and Paré, D. 2004. Productivity of black spruce and jack pine stands in Quebec as related to climate, site biological features and soil properties. *For. Ecol. Manage.* 191: 239–251.
- Houle, D., Bilodeau Gauthier, S., Paquet, S., Planas, D., and Warren, A. 2006. Identification of two genera of N_2 -fixing cyanobacteria growing on three feather moss species in boreal forests of Quebec, Canada. *Can. J. Bot.* 84: 1025–1029.
- Jacobs, D.F., Rose, R., Haase, D.L., and Alzugaray, P.O. 2004. Fertilization at planting impairs root system development and drought avoidance of Douglas-fir (*Pseudotsuga menziesii*) seedlings. *Ann. For. Sci.* 61: 643–651.
- Jobidon, R. 1995. Autécologie de quelques espèces de compétition d'importance pour la régénération forestière au Québec : revue de littérature. Mémoire de recherche forestière 117, Direction de la recherche forestière, Ministère des Ressources naturelles, Québec, Que.
- Kaspar, T.C., and Bland, W.L. 1992. Soil temperature and root growth. *Soil Sci.* 154: 290–299.
- Latham, R.E. 2003. Shrubland longevity and rare plant species in the northeastern United States. *For. Ecol. Manage.* 185: 21–39.
- Lavoie, M., and Bradley, R.L. 2003. Short-term increases in relative nitrification rates due to trenching in forest floor and mineral soil horizons of different forest types. *Plant Soil*, 252: 367–384.
- Littell, R.C., Milliken, G.A., Stroup, W.W., and Wolfinger, R.D. 1996. *SAS system for mixed models*. SAS Institute Inc., Cary, N.C.
- Louter, J.H., Kershaw, M., and Gordon, A.M. 1993. The autecology of competing non-crop vegetation on old field sites in southern Ontario. Science and Technology Transfer Unit (Southern Region), Ontario Ministry of Natural Resources, Brockville, Ont.
- Malik, V., and Timmer, V.R. 1996. Growth, nutrient dynamics, and interspecific competition of nutrient-loaded black spruce seedlings on a boreal mixedwood site. *Can. J. For. Res.* 26: 1651–1659.
- Mallik, A.U. 1995. Conversion of temperate forests into heaths: role of ecosystem disturbance and ericaceous plants. *Environ. Manage.* 19: 675–684.
- Mallik, A.U. 2001. Black spruce growth and understory species diversity with and without sheep laurel. *Agron. J.* 93: 92–98.
- Mallik, A.U. 2003. Conifer regeneration problems in boreal and temperate forests with ericaceous understory: role of disturbance, seedbed limitation, and keystone species change. *Crit. Rev. Plant Sci.* 22: 341–366.
- Margolis, H.A., and Brand, D.G. 1990. An ecophysiological basis for understanding plantation establishment. *Can. J. For. Res.* 20: 375–390.

- Marks, P.L. 1974. The role of pin cherry (*Prunus pensylvanica* L.) in the maintenance of stability in northern hardwood ecosystems. *Ecol. Monogr.* **44**: 73–88.
- Ministère des Ressources naturelles et de la Faune du Québec. 2005. Guide terrain : inventaire de qualification des plants résineux cultivés en récipients. Direction de la production des semences et des plants, Ministère des Ressources naturelles et de la Faune du Québec, Québec, Que.
- Morrison, I.K. 1974. Mineral nutrition of conifers with special reference to nutrient status interpretation: a review of literature. *Can. For. Serv. Publ.* 1343.
- Nilsson, M.-C., and Wardle, D.A. 2005. Understory vegetation as a forest ecosystem driver: evidence from the northern Swedish boreal forest. *Front. Ecol. Environ.* **3**: 421–428.
- Nordborg, F., and Nilsson, U. 2003. Growth, damage and net nitrogen uptake in *Picea abies* (L.) Karst. seedlings, effects of site preparation and fertilisation. *Ann. For. Sci.* **60**: 657–666.
- Näsholm, T., Ekblad, A., Nordin, A., Giesler, R., Höglberg, M., and Höglberg, P. 1998. Boreal forest plants take up organic nitrogen. *Nature (Lond.)*, **392**: 914–916.
- Örlander, G., Gemmel, P., and Hunt, J. 1990. Site preparation: a Swedish overview. FRDA Rep. 105, British Columbia Ministry of Forests, Victoria, B.C.
- Prescott, C., Weetman, G.F., and Barker, J.E. 1996. Causes and amelioration of nutrient deficiencies in cutovers of cedar–hemlock forests in coastal British Columbia. *For. Chron.* **72**: 293–302.
- Prévost, M., and Dumais, D. 2003. Croissance et statut nutritif de marcottes, de semis naturels et de plants d'épinette noire à la suite du scarifiage : résultats de 10 ans. *Can. J. For. Res.* **33**: 2097–2107.
- Rose, R., and Ketchum, J.S. 2003. Interaction of initial seedling diameter, fertilization and weed control on Douglas-fir growth over the first four years after planting. *Ann. For. Sci.* **60**: 625–635.
- Salifu, K.F., and Timmer, V.R. 2001. Nutrient retranslocation response of *Picea mariana* seedlings to nitrogen supply. *Soil Sci. Soc. Am. J.* **65**: 905–913.
- SAS Institute Inc. 2004. SAS/STAT® 9.1 user's guide. SAS Institute Inc., Cary, N.C.
- Saucier, J.P., Bergeron, J.F., Grondin, P., and Robitaille, A. 1998. Les régions écologiques du Québec méridional (3^e version). Supplément de l'Aubelle 124, Ministère des Ressources naturelles, Québec, Que.
- Schaalje, G.B., McBride, J.B., and Fellingham, G.W. 2001. Approximations to distributions of test statistics in complex mixed linear models using SAS® Proc MIXED. Pap. 262–26. In *Proceedings of the Twenty-Sixth Annual SAS® Users' Group International Conference*, Long Beach, Calif. SAS Institute Inc. Cary, N.C.
- Sen Tran, T., and Simard, R.R. 1993. Mehlich III-extractable elements. In *Soil sampling and methods of analysis*. Lewis Publishers, Boca Raton, Fla. pp. 43–49.
- Simpson, D.G., and Vyse, A. 1995. Planting stock performance: site and RGP effects. *For. Chron.* **71**: 739–742.
- Soil Classification Working Group. 1998. The Canadian system of soil classification. 3rd ed. Publication 1646, Agriculture and Agri-Food Canada, Ottawa, Ont.
- South, D., Rose, R., and McNabb, K. 2001. Nursery and site preparation interaction research in the United States. *New For.* **22**: 43–58.
- Swindel, B.F., Conde, L.F., and Smith, J.E. 1983. Plant cover and biomass response to clear-cutting, site preparation and planting in *Pinus elliotii* flatwoods. *Science (Wash., D.C.)*, **219**: 1421–1422.
- Swindel, B.F., Conde, L.F., and Smith, J.E. 1984. Species diversity: concept, measurement, and response to clearcutting and site-preparation. *For. Ecol. Manage.* **8**: 11–22.
- Thiffault, N., Cyr, G., Prigent, G., Jobidon, R., and Charette, L. 2004a. Régénération artificielle des pessières noires à éricacées : effets du scarifiage, de la fertilisation et du type de plants après 10 ans. *For. Chron.* **80**: 141–149.
- Thiffault, N., Titus, B.D., and Munson, A.D. 2004b. Black spruce seedlings in a *Kalmia-Vaccinium* association: microsite manipulation to explore interactions in the field. *Can. J. For. Res.* **34**: 1657–1668.
- Thiffault, N., Titus, B.D., and Munson, A.D. 2005. Silvicultural options to promote seedling establishment on *Kalmia-Vaccinium*-dominated sites. *Scand. J. For. Res.* **20**: 110–121.
- Timmer, V.R. 1991. Interpretation of seedling analysis and visual symptoms. In *Mineral nutrition of conifer seedlings*. CRC Press, Boston, Mass. pp. 113–134.
- Timmer, V.R., and Munson, A.D. 1991. Site-specific growth and nutrition of planted *Picea mariana* in the Ontario Clay Belt. IV. Nitrogen loading response. *Can. J. For. Res.* **21**: 1058–1065.
- Titus, B.D., Sidhu, S.S., and Mallik, A.U. 1995. A summary of some studies on *Kalmia angustifolia* L.: a problem species in Newfoundland forestry. Inf. Rep. N-X-296, Canadian Forest Service, St. John's, Nfld.
- Ung, C.-H., and Végiard, S. 1988. Problèmes d'inférence statistique reliés à la transformation logarithmique en régression. *Can. J. For. Res.* **18**: 733–738.
- van Den Driessche, R. 1991. Effects of nutrients on stock performance in the forest. In *Mineral nutrition of conifer seedlings*. CRC Press, Boston, Mass. pp. 229–260.
- Walinga, I., van der Lee, J.J., Houba, V.J.G., van Vark, W., and Novozamsky, I. 1995. Plant analysis manual. Kluwer Academic, Dordrecht, the Netherlands [looseleaf].
- Wall, R.E. 1977. Ericaceous ground cover on cutover sites in south western Nova Scotia. *Can. For. Serv. Inf. Rep. N-X-71*.
- Yamasaki, S.H., Fyles, J.W., Egger, K.N., and Titus, B.D. 1998. The effect of *Kalmia angustifolia* on the growth, nutrition, and ectomycorrhizal symbiont community of black spruce. *For. Ecol. Manage.* **105**: 197–207.
- Yamasaki, S.H., Fyles, J.W., and Titus, B.D. 2002. Interactions among *Kalmia angustifolia*, soil characteristics, and the growth and nutrition of black spruce seedlings in two boreal Newfoundland plantations of contrasting fertility. *Can. J. For. Res.* **32**: 2215–2224.
- Zhu, H., and Mallik, A.U. 1994. Interactions between *Kalmia* and black spruce: isolation and identification of allelopathic compounds. *J. Chem. Ecol.* **20**: 407–421.